ADAPTIVE TIME-LOCKED WIRELESS CHARGING STATION

A PROJECT REPORT

of

ECB1303 – EMBEDDED SYSTEMS & IOT DESIGN

Submitted by

NIRANJAN S	**(8115U24EC067)**
PRANAV M	**(8115U24EC075)**
SASI DHARAN M	**(8115U24EC100)**
VISHNU PRASATH S	**(8115U24EC121)**

in partial fulfilment for the award of the degree of

BACHELOR OF ENGINEERING
in
ELECTRONICS AND COMMUNICATION ENGINEERING

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
(AUTONOMOUS)
SAMAYAPURAM, TRICHY

ANNA UNIVERSITY
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Under the Guidance of
Dr. R. SAMSON DANIEL, M.E., Ph.D

Department of Electronics and Communication Engineering

K. RAMAKRISHNAN COLLEGE OF ENGINEERING
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ANNA UNIVERSITY, CHENNAI

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(AUTONOMOUS)
Under
ANNA UNIVERSITY, CHENNAI

BONAFIDE CERTIFICATE

Certified that this project report titled "**ADAPTIVE TIME-LOCKED WIRELESS CHARGING STATION**" is the bonafide work of **NIRANJAN S (8115U24EC067), PRANAV M (8115U24EC075), SASI DHARAN M(8115U24EC100), VISHNU PRASATH S (8115U24EC121)** who carried out the project work under my supervision. Certified further, that to the best of my knowledge the work reported here in does not form part of any other project report or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

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DECLARATION BY THE CANDIDATES

We declare that to the best of our knowledge the work reported herein has been composed solely by ourselves and that it has not been in whole or in part in any previous application for a degree.

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SIGNATURE OF THE CANDIDATES

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DEPARTMENT OF ECE

VISION

To be distinguished as a prominent program in Electronics and Communication Engineering Studies by preparing students for Industrial Competitiveness and Societal Challenges.

MISSION

M1. To equip the students with latest technical, analytical and practical knowledge

M2. To provide vibrant academic environment and Innovative Research culture

M3. To provide opportunities for students to get Industrial Skills and Internships to meet out the challenges of the society.

PROGRAM EDUCATIONAL OBJECTIVES (PEO'S)

PEO1: Graduates will become experts in providing solution for the Engineering problems in Industries, Government and other organizations where they are employed.

PEO2: Graduates will provide innovative ideas and management skills to enhance the standards of the society by individual and with team works through the acquired Engineering knowledge.

PEO3: Graduates will be successful professionals through lifelong learning and contribute to the society technically and professionally.

PROGRAM SPECIFIC OUTCOMES (PSO'S):

PSO1: Analyze and design complex electronic circuits and communication systems by applying the principles of engineering and mathematics.

PSO2: Analyze and provide innovative design solutions to research problems, industrial problems and for the demanding needs of the society in the domains of electronic integrated circuits, signal processing, high frequency and antenna design.

PSO3: Continuously involve in life-long learning activities so as to evolve and become an active team member in diverse and challenging design situations.

PROGRAM OUTCOMES:

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Engineering tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and the World: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of engineering practice.

PO8: Individual and Collaborative Teamwork: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO9: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large. Some of them are, being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO10: Project Management and Finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO11: Lifelong learning: Recognize the need for and have the preparation and ability to engage in independent and lifelong learning in the broadest context of technological change.

CO-PO MAPPING FOR
EMBEDDED SYSTEMS AND IOT DESIGN

After successful completion of this course, the students should be able to

CO1: Analyze the internal architecture and modelling of an embedded processor using design tools.

CO2: Develop the embedded system networks using different protocols.

CO3: Identify the evolution of the Internet of Things (IoT) using internal components for networking.

CO4: Evaluate cloud and big data analytics using IoT Platform.

	POs											PSOs			
COs	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3	Average
CO1	2.60	1.40	2.00	1.80	1.80	3.00	1.40	1.80	1.40	2.00	3.00	1.80	1.40	0.80	1.87
CO2	2.60	1.20	2.00	1.20	2.00	3.00	1.20	1.40	1.60	1.80	3.00	1.80	1.80	0.20	1.77
CO3	2.60	1.20	2.00	1.60	1.00	3.00	1.20	1.60	1.80	1.20	3.00	1.80	1.80	0.80	1.76
CO4	2.60	1.20	1.20	2.00	1.20	3.00	1.20	1.80	1.60	1.20	3.00	1.80	1.80	1.20	1.77
CO5	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00	3.00
Average	2.68	1.60	2.04	1.92	1.80	3.00	1.60	1.92	1.88	1.84	3.00	2.04	1.96	1.20	2.03

SUSTAINABLE DEVELOPMENT GOALS (SDGs)

Identify and select the Sustainable Development Goals (SDGs) that align with the objectives and outcomes of the project.

☐ Goal 1: No Poverty – End poverty in all forms everywhere

☐ Goal 2: Zero Hunger – End hunger and improve nutrition

☐ Goal 3: Good Health and Well-being – Ensure healthy lives

☐ Goal 4: Quality Education – Provide inclusive, equitable education

☐ Goal 5: Gender Equality – Empower women and girls

☐ Goal 6: Clean Water and Sanitation – Ensure safe water and sanitation

☐ Goal 7: Affordable and Clean Energy – Access to sustainable energy

☐ Goal 8: Decent Work and Economic Growth – Promote jobs and growth

☑ Goal 9: Industry, Innovation and Infrastructure – Build resilient infrastructure

☐ Goal 10: Reduced Inequalities – Reduce inequality

☑ Goal 11: Sustainable Cities and Communities – Make cities safe and sustainable

☑ Goal 12: Responsible Consumption and Production – Reduce waste

☐ Goal 13: Climate Action – Combat climate change

☐ Goal 14: Life Below Water – Protect oceans

☐ Goal 15: Life on Land – Protect ecosystems and biodiversity

☐ Goal 16: Peace, Justice and Strong Institutions – Promote justice and peace

☐ Goal 17: Partnerships for the Goals – Strengthen global partnerships

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ABSTRACT

Public places such as libraries, cafés, airports, shopping malls and educational institutions need safe and convenient smartphone charging facilities. However, existing public charging stations often lack proper security and control over charging time. To overcome these problems, this project proposes an Adaptive Time-Locked Wireless Charging Station that provides wireless charging, phone detection, password protection and timer-based operation. The system uses an ESP8266 NodeMCU as its main controller, along with a sound sensor, 4×4 keypad, Mini Servo SG90, relay module, 16×2 LCD display, buzzer and wireless charging module. The sound sensor detects whether a smartphone is placed on or removed from the charging platform.

Users can enter the charging time and security password using the keypad. The controller checks the sensor readings and user inputs before starting the charging process. The servo motor locks the compartment, while the relay switches ON the wireless charging module. The LCD displays instructions and charging status. A countdown timer controls the charging duration, and the relay automatically switches OFF when the selected time ends. The compartment can be opened only by entering the correct password. If an incorrect password is entered, the buzzer gives an alert and prevents unauthorized access. The system can be simulated using Wokwi for supported components, while physical testing is needed to verify the sensor and wireless charging performance. This project provides a simple and affordable solution for secure smartphone charging in public places. Future improvements may include IoT monitoring, advanced security and multiple charging compartments.

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CHAPTER-1

INTRODUCTION

1.1. INTRODUCTION

The increasing use of smartphones has created a growing need for convenient, reliable, and secure charging facilities in public places such as colleges, hospitals, airports, shopping malls, railway stations, libraries, bus terminals, and other common areas. Smartphones have become an essential part of daily life and are widely used for communication, education, navigation, online services, digital payments, entertainment, and professional activities. As a result, maintaining sufficient battery power throughout the day has become increasingly important. Users may spend several hours away from home and may not always have access to a suitable charger, charging cable, or electrical outlet. Public charging stations provide a useful solution by allowing users to recharge their smartphones whenever required. However, conventional public charging facilities generally concentrate only on supplying electrical power and provide limited protection for the user's device.

In a typical public charging arrangement, the smartphone remains exposed on a table or charging point while the charging process takes place. The user may therefore have to remain near the charging station to prevent accidental or unauthorized handling of the device. This can be inconvenient when the user needs to attend another activity. In addition, shared charging points may create difficulties in identifying which device belongs to which user when several people use the same facility. Therefore, a public charging system that combines charging convenience with controlled physical access can provide a more organized solution.

Wireless charging provides an alternative to conventional wired charging by eliminating the need for a direct charging cable. A compatible smartphone can be placed on a wireless charging module, allowing power to be transferred without physically connecting a cable to the device. This improves convenience and reduces cable-related problems. However, a conventional open wireless charging arrangement still leaves the smartphone physically accessible during charging. Wireless

charging alone therefore does not address the security requirements of a public charging environment.

The proposed Adaptive Time-Locked Wireless Charging Station addresses the limitations of conventional public charging facilities by combining wireless charging with controlled access and time-based operation. The system uses an ESP8266 NodeMCU, a Buzzer with an SOUND interface, a servo-based locking mechanism, a 4×4 matrix keypad, an LCD, a relay and a buzzer. The Buzzer detects the placement of the smartphone by measuring the change in Buzzer applied to the charging platform. The servo mechanism controls access to the charging compartment.

When a smartphone is placed on the platform and the required charging information is entered, the controller coordinates the locking mechanism and charging operation. The relay supplies power to the wireless charging module for the selected duration. The LCD displays the operating status, and the buzzer provides an alert when an incorrect authentication code is entered. When the timer expires, charging stops, and the compartment remains locked until the correct security code is entered.

Once the charging session begins, the relay activates the wireless charging module and the compartment is locked. The selected charging duration is controlled by the programmed timer. The LCD provides information and instructions to the user during the different stages of operation. When the selected charging period expires, the relay switches OFF and the charging operation stops. The user can then enter the correct authentication code to unlock the compartment and retrieve the smartphone.

The proposed system also incorporates a Buzzer-sound-based detection mechanism for identifying the placement of the smartphone inside the charging compartment. The Buzzer measures the Buzzer applied to the charging platform and provides an indication when the smartphone is placed in the designated position. This approach replaces the previously considered IR-based detection method and provides a alert-based method for confirming device placement. The Buzzer measurement can be processed by the ESP8266 controller and used as an input for the charging and locking sequence. Thus, the Buzzer becomes an important sensing element that links the physical placement of the smartphone with the automated operation of the charging station

1.2. EXISTING WORK

Existing public charging systems generally provide electrical power to users through wired USB ports, wall adapters, charging cables or open wireless charging pads. These systems are commonly installed in public locations where users may need to recharge their smartphones while travelling, studying, shopping or waiting. The main purpose of these systems is to make electrical power easily accessible without requiring the user to carry a personal power source at all times.

In a conventional wired charging arrangement, the user connects the smartphone to the charging station using an appropriate cable. Although this method is straightforward, different smartphones may require different types of connectors. The availability of a compatible cable can therefore become inconvenient, particularly in shared public charging facilities. Cable damage, loose connections and improper handling can also affect the charging process.

Wireless charging pads provide a more convenient alternative because they eliminate the need for a physical cable connection. The user simply places a compatible smartphone on the charging surface. However, the device generally remains exposed on the charging pad throughout the charging period. In a public environment, the user may have to remain close to the station to protect the smartphone from accidental movement or unauthorized handling.

Some charging kiosks additionally provide timers, payment mechanisms or electronic locking systems. These systems demonstrate that charging duration and access can be controlled electronically. However, such functions may be implemented separately and may not always be combined into a simple, low-cost embedded system.

Therefore, existing systems mainly focus on providing charging power and convenience, while physical protection, controlled charging duration and access management may remain limited.

1.3. DRAWBACKS FOUND IN EXISTING WORK

1. Lack of Physical Security: Smartphones remain exposed on open charging pads, making them vulnerable to theft or unauthorized access.

2. Continuous User Monitoring: Users may need to stay near the charging station to ensure the safety of their smartphones.

3. Cable Dependency: Wired charging systems require compatible charging cables, which may not always be available.

4. Cable-Related Issues: Damaged, loose or missing cables reduce the convenience and reliability of charging systems.

5. Manual Connection: Wired charging requires users to physically connect and disconnect their smartphones.

6. Security Issues in Open Wireless Charging: Open wireless charging pads do not provide physical protection, making users uncomfortable leaving their devices unattended.

7. Lack of Time-Based Access Control: Basic charging systems generally do not provide automatic control over the charging duration.

8. Limited Integration of Security Features: Some advanced charging kiosks provide timers, payment systems and electronic locks, but these features are not always integrated into a single system.

9. High Cost of Advanced Systems: Advanced charging kiosks with integrated security and access-control features may increase the overall system cost.

10. Need for an Integrated Solution: Existing systems lack a combination of wireless charging, physical security, time-based control and convenient operation.

1.4. NEED FOR THE PROPOSED WORK

The need for the proposed work arises from the limitations associated with conventional public charging systems. Smartphones are frequently used in public places, and their battery level may become low when users are away from home. Although charging points are available in many locations, users may hesitate to leave their smartphones unattended because conventional charging arrangements normally do not provide adequate physical protection.

A suitable public charging system should therefore provide both convenient power transfer and

controlled access to the device. Wireless charging can eliminate the need for charging cables and make the charging process easier for users. However, wireless charging alone does not provide physical security. A protective compartment with an electronic locking mechanism can help address this issue by restricting access to the smartphone during the charging session.

Time control is another important requirement. The charging session can be initiated for a user-selected duration, allowing the system to control the charging period. Combining time control with a locking mechanism creates a more organized charging process. The proposed system also uses phone detection to determine whether the device is positioned inside the charging compartment before the charging operation proceeds.

Authentication provides an additional layer of access control. A keypad can be used to enter a security code and charging-duration information. The system can maintain the locked condition during the charging process and allow authorized access after the required operation is completed.

Thus, the proposed work is needed to integrate wireless charging, time control, phone detection, authentication and physical access control into a single embedded charging station. Such integration can make public charging more convenient and provide better protection for smartphones compared with a basic open charging arrangement.

The proposed Adaptive Time-Locked Wireless Charging Station can be applied in various public and semi-public locations where users require a secure place to charge their smartphones. One important application is in colleges and educational institutions, where students may spend several hours on campus and may require charging facilities between classes. The locked charging compartment can allow students to charge their smartphones without continuously staying beside the charging station.

The system can also be used in shopping malls and commercial centres, where customers may remain at the location for extended periods. A dedicated charging station can provide convenient wireless charging while reducing the need for users to carry charging cables. Similarly, the system can be installed in airports and railway stations, where passengers often have waiting periods and may need to recharge their smartphones before continuing their journey.

Another possible application is in hospitals and healthcare facilities, where visitors and staff may

require temporary smartphone charging facilities. The controlled charging arrangement can provide a designated location for charging while helping protect the device.

The proposed system may also be suitable for libraries, offices, hotels, restaurants, cafés and transportation hubs. In these environments, users can place their compatible smartphone inside the charging compartment, enter the required information through the keypad and allow the system to control the charging operation.

The combination of wireless charging, phone detection, time-based control, authentication and a servo-operated locking mechanism makes the system suitable for locations where both convenience and controlled access are desirable.

CHAPTER-2

LITERATURE SURVEY

2.1. LITERATURE SURVEY

1. IoT Wireless EV Charging with Real-Time Energy Monitoring – Zenodo, 2026 (International Journal)

Key Findings:

- Integrates Wireless Power Transfer (WPT) with a real-time IoT monitoring dashboard.
- Provides monitoring of the wireless charging process through an IoT-based system.
- Demonstrates the use of IoT technology for monitoring charging operations.

Challenges:

- Adds network dependency to the charging system.
- Increases the overall system complexity.

2. IoT-Based EV Charging Station System (WSN + BLE) – Bhavana P., 2023

Key Findings:

- Uses a Firebase backend for the charging system.
- Uses a BLE-based sensor network for communication.
- Provides an Android monitoring application.
- Combines wireless sensing and IoT monitoring for charging-station operation.

Challenges:

- Slot availability remains inaccurate.
- Charge-time prediction remains inaccurate.
- The use of multiple communication technologies can increase system complexity.

3. TDMA in Adaptive Resonant Beam Charging – Xiong et al. (IEEE) - 2019

Key Findings:

- Uses Time-Division Multiple Access (TDMA) scheduling.
- Supports concurrent multi-device Wireless Power Transfer (WPT).
- Allocates charging access among multiple devices using time-based scheduling.

Challenges:

- Scheduling overhead increases with the number of connected devices.
- Managing multiple devices makes the charging system more complex.

4. Adaptive Control of IoT/M2M in Smart Buildings – Djehaiche et al. (IEEE) - 2023

Key Findings:

- Integrates a heterogeneous Wireless Sensor Network (WSN) with a mobile soundular network.
- Provides device control using different wireless communication technologies.
- Demonstrates IoT/M2M-based control for smart environments.

Challenges:

- Interoperability between multiple wireless protocols is difficult.
- Combining different wireless protocols increases system cost.

5. Smart & Secured Mobile Charging Station (Public Place) – Md. Eftekhar Alam - 2021

Key Findings:

- Uses a coin-recognition module to set the charging duration.
- Uses an RFID-controlled solenoid door lock.
- Combines charging-duration control with physical access control.
- Provides a security mechanism for public charging applications.

Challenges:

- Fixed coin-based timing is inflexible.
- Public charging kiosks remain relatively security-weak.
- The charging duration is not dynamically related to the actual charging condition.

6. IoT-Based Wireless EV Charging Station (IR-Triggered) – Honnesh Mandapati (IEEE) - 2021

Key Findings:

- Uses an IR sensor to automatically activate charging mode.
- Provides a mobile application for monitoring and location-related functions.
- Demonstrates automatic charging activation using device detection.

- The IR-based detection concept is relevant to automatic detection before charging.

Challenges:

- Reliability varies according to device positioning.
- Environmental conditions can affect detection reliability.
- The system does not provide the complete physical time-locking mechanism of the proposed system.

7. Foreign Object Detection in WPT Systems (Review) – J. Lu, G. Zhu (IEEE) - 2022

Key Findings:

- Analyses magnetic-flux distortion caused by metallic objects.
- Considers foreign-object detection under the Qi standard.
- Focuses on improving safety in inductive wireless power-transfer systems.

Challenges:

- Detection accuracy depends heavily on coil geometry.
- Detection performance is affected by the coupling between the charging coils.

8. Foreign Object Detection Testing for Wireless Chargers – USPTO Patent - 2023

Key Findings:

- Uses power-dissipation comparison between transmitted and received power.
- Differences in power can be used to identify unwanted objects.
- Provides a method for improving wireless-charging safety.

Challenges:

- Coupling misalignment can trigger false-positive detections.
- Detection accuracy depends on the coupling condition between the transmitter and receiver.

9. Advances in EV Wireless Charging Technology – A Systematic Review and Future Trends – B. Latha, Mohammed Mujahid Irfan, Aymen Flah - 2024

Key Findings:

- Reviews the evolution of the Qi wireless-charging standard.
- Covers Foreign Object Detection (FOD).

- Discusses transmitter authentication.
- Reviews wireless power-transfer technologies and their development.

Challenges:

- Higher power delivery increases transmitter safety-circuit complexity.
- Advanced wireless-charging systems require more complex safety mechanisms.

CHAPTER-3

PROJECT DESCRIPTION

3.1 PROPOSED SYSTEM

The proposed Adaptive Time-Locked Wireless Charging Station is designed to provide secure and time-controlled wireless charging for smartphones. The system uses an ESP8266 NodeMCU as its central controller and integrates a Mini Servo SG90, Buzzer with SOUND interface, 4×4 matrix keypad, 16×2 LCD display, relay module, wireless charging module and buzzer.

The Buzzer is positioned beneath the charging platform to measure changes in the applied Buzzer. Its output is processed through the SOUND interface, allowing the ESP8266 to determine whether a smartphone has been placed on or removed from the platform. Appropriate calibration and a detection threshold are required to distinguish the occupied and unoccupied conditions.

The 4×4 matrix keypad allows the user to enter the charging duration and security code. The ESP8266 processes these inputs and controls the servo motor, relay, LCD and buzzer according to the programmed sequence. The servo motor operates the compartment's locking mechanism, while the relay controls the power supplied to the wireless charging module.

When a charging session is initiated and the smartphone is detected, the system locks the compartment and activates the charging operation according to the selected duration. The LCD displays instructions and status information. When the countdown expires, the relay switches OFF. The compartment remains locked until the correct authentication code is entered. An incorrect code activates the buzzer and prevents unlocking.

After the user retrieves the smartphone, the Buzzer-sound reading returns towards the unBuzzered reference value. The controller can use this change to identify phone removal and return the system to standby. The ESP8266 also provides Wi-Fi capability for possible future IoT-based monitoring, although network connectivity is not required for the basic operation of the prototype.

3.2 FLOW DIAGRAM

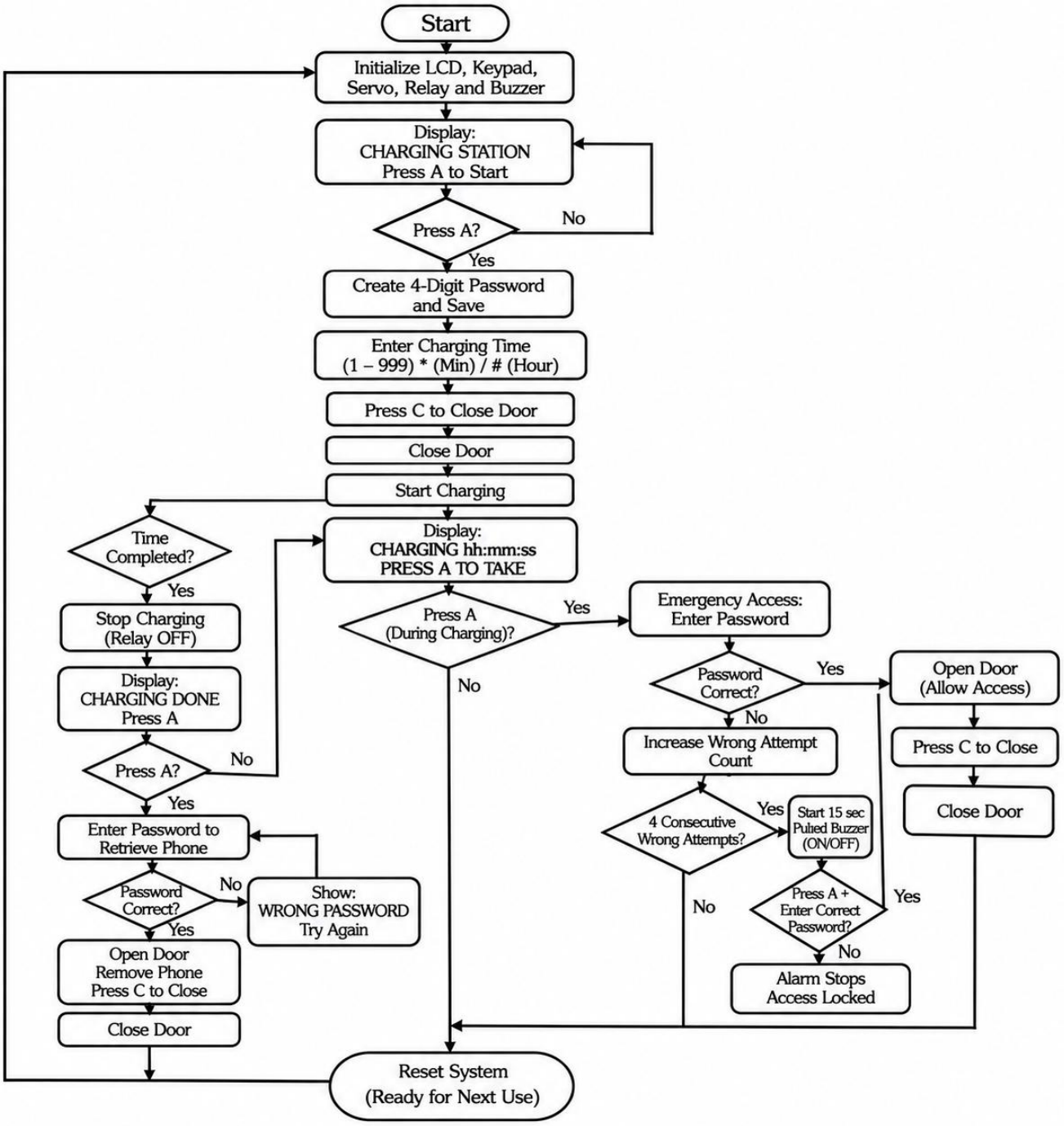


Figure 3.1: Flow Diagram of Adaptive Time-Locked Wireless Charging Station

3.3 COMPONENTS USED

The proposed system uses several electronic and electromechanical components, with each component performing a specific function in the charging, sensing, authentication, display or locking operation.

Table 3.1: Component List of the Adaptive Time-Locked Wireless Charging Station

S.No	Component	Specification	Quantity
1	ESP8266 Microcontroller	NodeMCU v1.0 / v2	1
2	Servo Motor	Mini Servo SG90 (5 V)	1
3	Relay Module	5 V Single-Channel	1
4	4×4 Matrix Keypad	Membrane type	1
5	LCD Display	16×2 with I2C module	1
6	Power Adapter	9 V / 2 A DC	1
7	Breadboard	Mini	1
8	Connecting Wires, Female Power Port	Male-Male, Male-Female, DC Jack (5.5 mm x 2.1 mm)	As needed
9	Wireless coil	5v coil	1
10	Buzzer	buzzer(-)	1

1**. ESP8266 NodeMCU**

The ESP8266 NodeMCU is the main microcontroller of the system. It controls the complete operation of the charging station. It scans the keypad, reads the IR sensor, controls the servo motor and relay, communicates with the LCD and manages the charging countdown timer.

2. Mini Servo SG90

The SG90 servo motor is used for operating the charging-compartment door. It provides the physical locking and unlocking mechanism. The servo changes its position according to commands from the ESP8266.

3.**Buzzer with SOUND Module**

The SOUND is used as the interface between the Buzzer and ESP8266. It amplifies the very small electrical signal produced by the Buzzer and converts it into a digital value that can be processed by the

controller. The module allows the system to monitor changes in Buzzer and determine whether the smartphone has been placed on or removed from the charging platform.

4. 4x4 Matrix Keypad

The 4×4 matrix keypad provides the user-input interface. It allows the user to enter the required charging duration and security code. The ESP8266 processes these inputs and uses the stored authentication code to determine whether the compartment can be unlocked after charging.

5. 16×2 LCD with I2C

The LCD provides instructions and status information to the user. The I2C interface reduces the number of GPIO connections required between the ESP8266 and the display.

6. 5 V Single-Channel Relay

The relay operates as an electronic switch for the wireless charging coil. It is activated when charging starts and deactivated when the charging timer expires.

7. Wireless Charging Coil

The wireless charging coil transfers electrical energy to a compatible smartphone without requiring a direct charging cable. It is the main charging element of the physical prototype.

8. 9 V / 2 A DC Adapter

The DC adapter provides the required power for the prototype circuit and charging system.

9. Breadboard

The breadboard is used to assemble and interconnect the electronic components during prototype development.

10. DC Jack and Connecting Wires

The DC jack provides the power connection, while male-male and male-female connecting wires are used to establish electrical connections between the modules.

3.4 CIRCUIT DIAGRAM AND ITS EXPLANATION

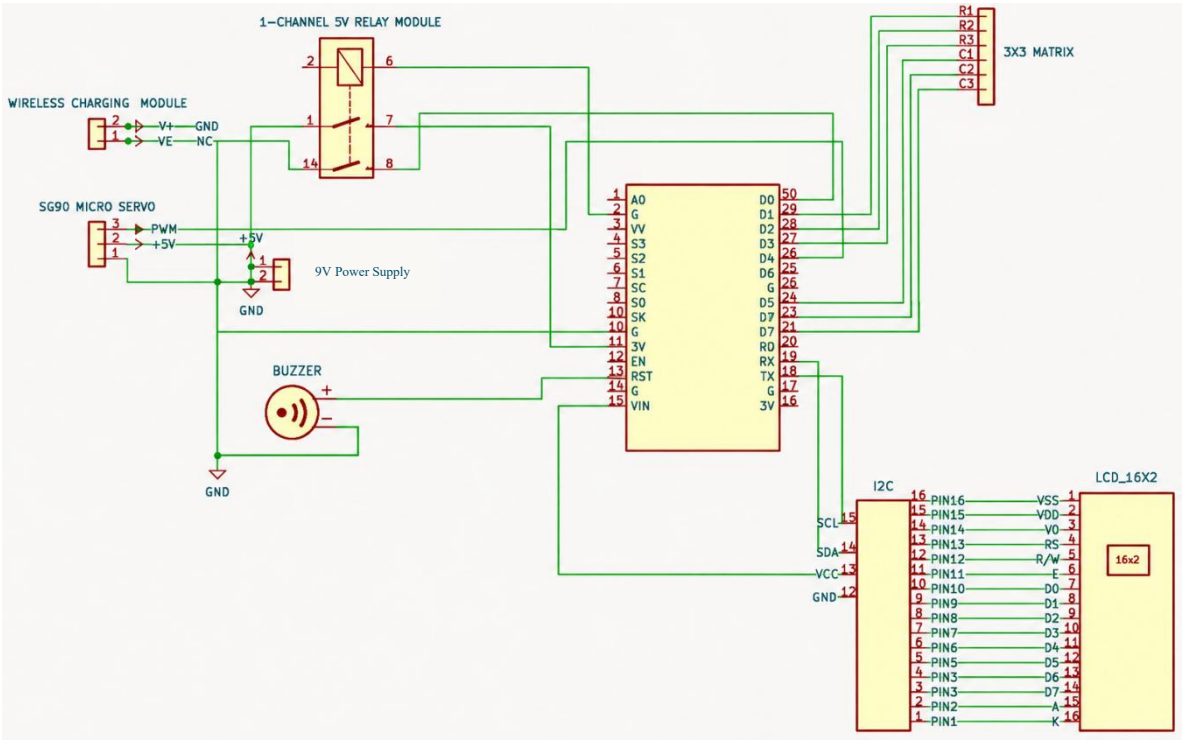


Figure 3.2: Circuit diagram of the Adaptive Time-Locked Wireless Charging Station

1. ESP8266 NodeMCU:

The ESP8266 NodeMCU is the central controller responsible for processing sensor readings, reading keypad inputs, controlling the charging timer and operating the output devices. The board provides digital GPIO pins for communication with the SOUND module, keypad, servo motor, relay and buzzer. Its I2C interface is used for communication with the LCD display. The controller executes the program stored in its memory and coordinates the charging and security operations. The ESP8266 operates with 3.3 V GPIO logic, so external modules must not apply incompatible voltage levels to its input pins.

2. Buzzer and SOUND Interface:

The Buzzer is positioned beneath the smartphone charging platform to detect changes in the applied Buzzer. Its E+ and E− terminals connect to the excitation-positive and excitation-negative terminals of the SOUND module, respectively. Its A+ and A− terminals connect to the corresponding

signal inputs of the SOUND. The SOUND module processes the small electrical signal produced by the Buzzer and converts it into digital readings. The SOUND DT pin can be connected to D5 (GPIO14) of the ESP8266, while its SCK pin can be connected to D6 (GPIO12), provided these pins are available in the final circuit. The controller reads the Buzzer measurements and compares them with a calibrated reference value and detection threshold to determine whether the smartphone has been placed on or removed from the platform. The SOUND must be powered according to its module specifications, and its logic connections must be compatible with the ESP8266.

3. 4×4 Matrix Keypad:

The 4×4 matrix keypad contains four row connections, R1 to R4, and four column connections, C1 to C4. These eight signal lines allow the ESP8266 to identify the pressed key by scanning the rows and columns. The keypad is used to enter the required charging duration and security code. The program processes the entered values and stores the authentication code for subsequent verification. The keypad's eight signal connections must be assigned to eight available GPIO pins. Because the ESP8266 has a limited number of usable GPIOs, the keypad pin assignment must be coordinated with the SOUND, LCD, servo, relay and buzzer connections. If the available GPIOs are insufficient, an I/O expander may be used.

4. 16×2 LCD with I2C Interface:

The 16×2 LCD displays instructions, charging status, remaining time and authentication messages. Its VCC and GND pins connect to a suitable power supply and ground. The SDA pin carries serial data, while the SCL pin carries the clock signal. On a typical ESP8266 NodeMCU configuration, D2 (GPIO4) can be used for SDA and D1 (GPIO5) for SCL. These assignments must match the connections defined in the program. The I2C interface reduces the number of GPIO connections required by the display. The LCD module's supply voltage and I2C logic levels must be checked before connecting it to the ESP8266.

5. SG90 Servo Motor:

The SG90 servo motor controls the physical locking and unlocking mechanism of the charging compartment. It normally has three wires: power, ground and control signal. The control signal can be connected to an available GPIO pin, such as D0 (GPIO16), if supported by the selected servo-control implementation and if that pin is not used by another component. The servo's power wire must connect to a supply suitable for the motor, and its ground must share a reference with the ESP8266. The controller sends position commands to move the servo between the programmed locked and unlocked positions. A suitable external supply is recommended when the servo current could exceed the capability of the controller's supply.

6. Relay Module:

The relay module controls the power supplied to the wireless charging module. Its IN pin connects to an available ESP8266 GPIO output, while VCC and GND connect to the module's specified supply and ground. For example, D7 (GPIO13) may be used as the relay-control output if it is available in the final pin allocation. The relay contacts are connected in series with the appropriate power supply line for the wireless charging module. When charging begins, the ESP8266 activates the relay. When the programmed countdown reaches zero, the controller deactivates the relay to interrupt power to the charging module. The relay's contact ratings must be suitable for the switched voltage and current.

7. Wireless Charging Module:

The wireless charging module transfers energy to a compatible smartphone without requiring a direct charging cable. Its supply input is connected through the relay-controlled power circuit so that the charging operation can be started and stopped by the ESP8266. The module must receive the voltage and current specified by its manufacturer. Its power connections must not be connected directly to an ESP8266 GPIO pin. The physical charging performance must be evaluated using the actual charging module and a compatible smartphone.

8. Buzzer:

The buzzer provides an audible alert when the user enters an incorrect authentication code. Its control connection can be connected to an available GPIO output, such as D4 (GPIO2), provided that this pin is not assigned to another device and the buzzer can be driven safely by the chosen circuit. If the buzzer requires more current than the GPIO can supply, a suitable transistor driver must be used. The program activates the buzzer when authentication fails and keeps the compartment locked.

9. Power Supply and Ground Connections:

The power supply provides energy to the ESP8266 and the connected electronic modules. The ESP8266 requires a suitable regulated supply, while the servo motor, relay and wireless charging module must receive power appropriate to their specifications. The SOUND and LCD must also be supplied at compatible voltage levels. A common ground reference is required for modules exchanging control signals, unless the circuit uses an appropriately isolated interface. The power arrangement must prevent excessive current drawn, voltage drops and electrical noise from affecting the controller or sensor measurements.

3.5 SOFTWARE USED

The software development of the Adaptive Time-Locked Wireless Charging Station involves programming the ESP8266 NodeMCU, interfacing the electronic components, implementing the charging-control algorithm and verifying the program operation. The Arduino Integrated Development Environment (IDE) is used for writing, compiling and upBuzzering the program to the ESP8266 NodeMCU. The Wokwi online simulator is used to verify the supported control logic and component interactions before or during hardware implementation. Embedded C/C++ programming is used to develop the instructions required for sensor reading, keypad input, authentication, timer management and control of the connected devices.

3.5.1 Arduino IDE

The Arduino IDE is the primary software development environment used to program the ESP8266 NodeMCU. It provides a code editor for writing the program, a compiler for checking the code and a facility for upBuzzering the compiled program to the microcontroller through a USB connection. The ESP8266 board support package must be installed in the Arduino IDE to enable the selection and programming of the NodeMCU board.

To use the Arduino IDE, the ESP8266 board support package is installed through the Boards Manager. The appropriate ESP8266 NodeMCU board is then selected from the Tools menu, and the correct communication port is chosen after connecting the board to the computer using a USB cable. The program is written in the editor, compiled to identify syntax or compilation errors, and upBuzzered to the board. The Serial Monitor can be used to observe diagnostic messages, sensor readings and program status when these messages are included in the firmware.

3.5.2 Wokwi Online Simulation

Wokwi is an online electronics simulator that can be used to test supported microcontrollers, electronic components and program logic in a virtual environment. It allows the programmer to connect supported components, define their pin connections, Buzzer the program and observe the resulting behaviour without immediately assembling the entire physical circuit.

For this project, Wokwi can be used to examine supported control operations, such as keypad input, LCD messages, servo commands, relay switching and timer-controlled operation. The simulation is useful for identifying programming errors and checking the intended sequence of operations. However, the Buzzer, SOUND interface and wireless power-transfer operation must be tested using suitable simulation models if available or verified separately using the physical hardware. Simulation results must therefore be reported only for functions that were actually represented and tested.

3.6 WORKING PRINCIPLE

The working principle of the Adaptive Time-Locked Wireless Charging Station is based on Buzzer-sound-based smartphone detection, wireless charging, timer control, user authentication and physical locking.

Initially, the system remains in standby mode. The ESP8266 initializes the connected components, and the LCD displays the required instructions. The user initiates a charging session and enters the required charging duration and security code using the 4×4 matrix keypad. When the smartphone is placed on the charging platform, the Buzzer detects the change in applied Buzzer through the SOUND interface. The ESP8266 processes whether platform is occupied. After the required conditions are satisfied, the servo motor locks the compartment and the relay activates the wireless charging module.

The countdown timer starts for the selected charging duration. During this period, the compartment remains locked and the LCD provides status information. When the countdown reaches zero, the ESP8266 switches OFF the relay to stop the charging operation.

After the charging period ends, the user enters the previously stored security code. If the code is correct, the servo unlocks the compartment and permits retrieval of the smartphone. If the code is incorrect, the buzzer provides an alert and remains locked.

When the smartphone is removed, the Buzzer-sound reading changes towards the unBuzzered reference value. The controller can use this condition to return the system to standby and prepare for the next charging session.

3.6.1 ESP8266 BOARD SUPPORT AND EMBEDDED C/C++

The ESP8266 NodeMCU acts as the main processing unit of the charging station. Its firmware is written in Embedded C/C++ using the Arduino programming framework. The program begins with an initialization section that configures the input and output pins, initializes the LCD and sensor interface, attaches the servo-control mechanism and prepares the relay and buzzer for operation.

The main program continuously checks the operating conditions and executes the appropriate actions. It reads the keypad input, obtains Buzzer measurements through the SOUND interface, updates the LCD, manages the charging countdown and controls the servo motor and relay. Conditional statements are used to determine whether the platform is occupied, whether the charging duration has expired and whether the entered authentication code is correct. This allows the controller to coordinate sensing, charging and physical security through a single program.

3.6.2 BUZZER AND SOUND SOFTWARE INTERFACE

The Buzzer is used to identify changes in the Buzzer applied to the smartphone charging platform. Since the Buzzer produces a small electrical signal, the SOUND interface module is used to process this signal and provide digital measurements to the ESP8266. An appropriate SOUND software library is included in the program to communicate with the interface module.

During initialization, the program configures the SOUND data and clock pins. The Buzzer is calibrated to establish a reference reading for the empty platform. When a smartphone is placed on the platform, the measured Buzzer changes. The program compares the reading with a predefined detection threshold to determine whether the platform is occupied. A corresponding change in the reading can be used to detect phone removal. Calibration and threshold selection are important because sensor readings may vary due to mechanical placement, vibration and measurement noise.

3.6.3 MATRIX KEYPAD PROGRAMMING

The 4×4 matrix keypad provides the user interface for entering the charging duration and authentication code. The keypad consists of four rows and four columns, allowing sixteen keys to be identified using eight signal connections. A suitable keypad library can be used to scan the row and column connections and identify the pressed key.

The program reads the user's input and interprets the selected keys according to the programmed procedure. The entered charging duration is stored for use by the countdown timer, and the security code is stored for subsequent authentication. After the charging period ends, the user enters the code again. The program compares the entered code with the stored code. If both match, the controller permits the servo motor to unlock the compartment. If they do not match, the compartment remains locked and the buzzer is activated.

3.6.4 LCD PROGRAMMING USING I2C

The 16×2 LCD with an I2C interface is used to display operating instructions, charging duration, system status and authentication messages. The I2C interface reduces the number of GPIO pins required for communication between the LCD and the ESP8266. A compatible LCD library is included in the program to initialize the display, position the cursor and print messages.

During system initialization, the LCD displays a starting message or standby instruction. During operation, it can display keypad instructions, the selected charging duration, the charging status and the remaining time. After the timer expires, it can display a message indicating that charging has ended and authentication is required. If an incorrect code is entered, the display can show an access-denied message.

3.6.5 SERVO MOTOR CONTROL

The Mini Servo SG90 is controlled by sending position commands from the ESP8266. A compatible servo-control library can be used to initialize the control pin and specify the required servo positions. The program defines the positions corresponding to the unlocked and locked states of the compartment, based on the actual mechanical arrangement.

When a charging session is initiated, the servo can move to the unlocked position to allow the smartphone to be placed inside the compartment. Once the required conditions are satisfied, the

controller commands the servo to the locked position. After the timer expires, the compartment remains locked until successful authentication. When the correct code is entered, the servo moves to the unlocking position and allows the user to retrieve the smartphone.

3.6.6 RELAY AND CHARGING TIMER PROGRAMMING

The relay module is controlled through a digital output of the ESP8266. The program defines the relay-control pin and determines the correct ON and OFF output states according to the relay module's electrical configuration. When a charging session begins, the relay activates the supply to the wireless charging module. When the selected charging duration expires, the controller deactivates the relay to stop power delivery.

The charging timer is implemented in the program using the duration entered through the keypad. The controller records the starting time and calculates the remaining duration. A non-blocking timing method, such as using the millis() function, can be used to update the countdown without unnecessarily stopping keypad scanning, sensor reading or other control tasks. When the countdown reaches zero, the program switches off the relay and updates the LCD. The actual charging duration depends on the value entered by the user and the timer implementation.

3.6.7 BUZZER AND AUTHENTICATION LOGIC

The buzzer provides an audible alert when the authentication procedure fails. The program compares the code entered through the keypad with the code stored during the charging session. When the entered code is correct, the controller commands the servo to unlock the compartment. When the entered code is incorrect, the controller maintains the locked state and activates the buzzer for the programmed interval.

The buzzer operation is controlled through a suitable GPIO output, using a driver circuit if required by the buzzer's current requirements. The LCD can also display the result of the authentication attempt. These functions provide both audible and visual feedback to the user.

3.7 ADVANTAGES

1. Physical Security:

The smartphone is kept inside a locked compartment during the charging process.

2. Wireless Charging:

A compatible smartphone can be charged without directly connecting a charging cable.

3. Time-Controlled Charging:

The relay automatically terminates the charging operation when the selected time expires.

4. User Authentication:

The security code helps ensure that only the authorized user can retrieve the phone.

5. Automatic Phone Detection:

The IR sensor detects the presence of the phone inside the compartment.

6. User-Friendly Operation:

The keypad and LCD provide a simple interface for entering information and receiving system instructions.

7. Low-Cost Implementation:

The system uses commonly available embedded-system components.

8. Simulation Before Hardware:

Wokwi allows the control logic and major circuit connections to be checked before physical implementation.

9. Expandability:

The ESP8266's built-in Wi-Fi capability provides the possibility of future IoT-based monitoring

and remote-control features.

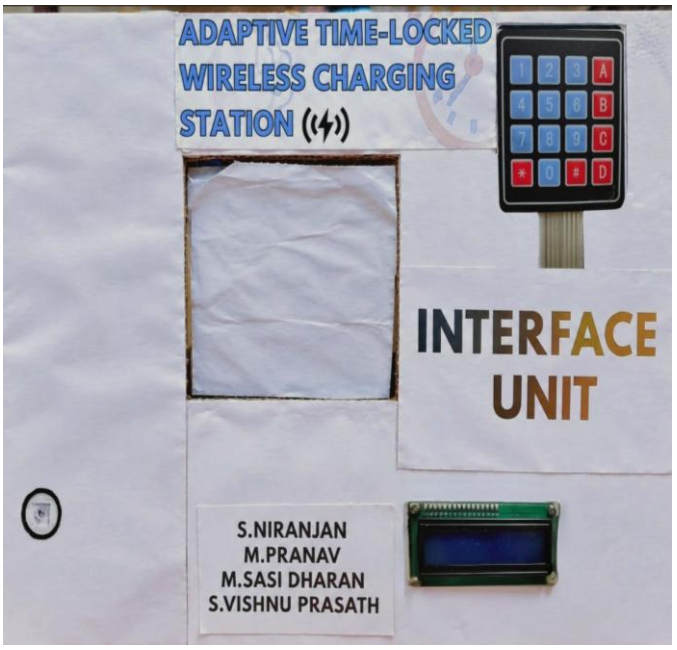
10. Controlled Power Usage:

The relay ensures that the wireless charging coil receives power only during the active charging period.

CHAPTER-4

RESULTS AND DISCUSSION

4.1 PROTOTYPE



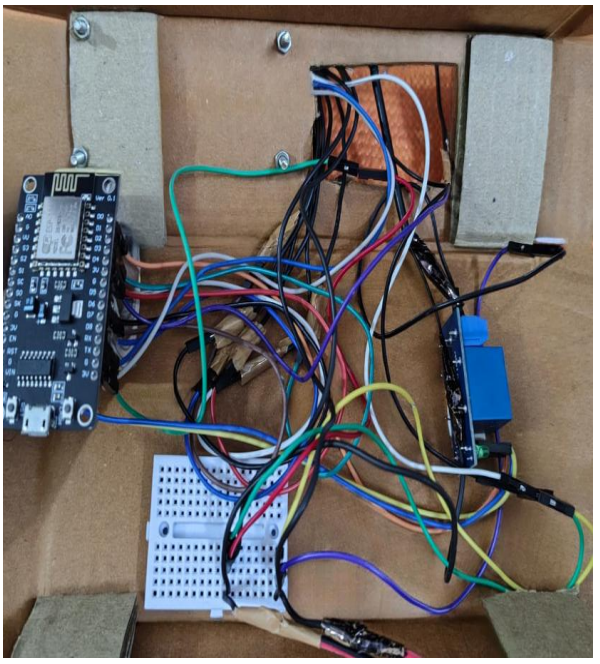


Figure 4.1: Interface of the Adaptive Time-Locked Wireless Charging Station

Figure 4.2: Internal Control of the Adaptive Time-Locked Wireless Charging Station

The prototype of the Adaptive Time-Locked Wireless Charging Station integrates the ESP8266 NodeMCU with a Buzzer and SOUND interface, 4×4 matrix keypad, 16×2 LCD with I2C interface, Mini Servo SG90, relay module, wireless charging module and buzzer. The objective of the prototype is to demonstrate the integration of wireless charging, Buzzer-based smartphone detection, user authentication, time control and physical access protection.

The ESP8266 NodeMCU acts as the main control unit. It processes keypad inputs, reads the Buzzer-sound measurements through the SOUND interface and controls the servo motor, relay, LCD and buzzer according to the programmed sequence.

When the smartphone is placed on the charging platform, the Buzzer measures the change in applied Buzzer. The controller uses the processed reading to determine whether the platform is occupied. The servo motor operates the compartment lock, while the relay controls power to the wireless charging module.

The user enters the charging duration and authentication code using the 4×4 matrix keypad. The LCD displays instructions and operating status. When the selected charging duration expires, the relay switches OFF. Correct authentication allows the compartment to be unlocked, while an incorrect code activates the buzzer and maintains the locked condition.

The prototype should be evaluated for reliable Buzzer detection, correct keypad operation, timer-controlled relay switching, authentication, servo movement and return to standby after phone removal. Any reported response times or test outcomes should be based on actual measurements obtained during prototype testing.

4.2 SIMULATION OUTPUT

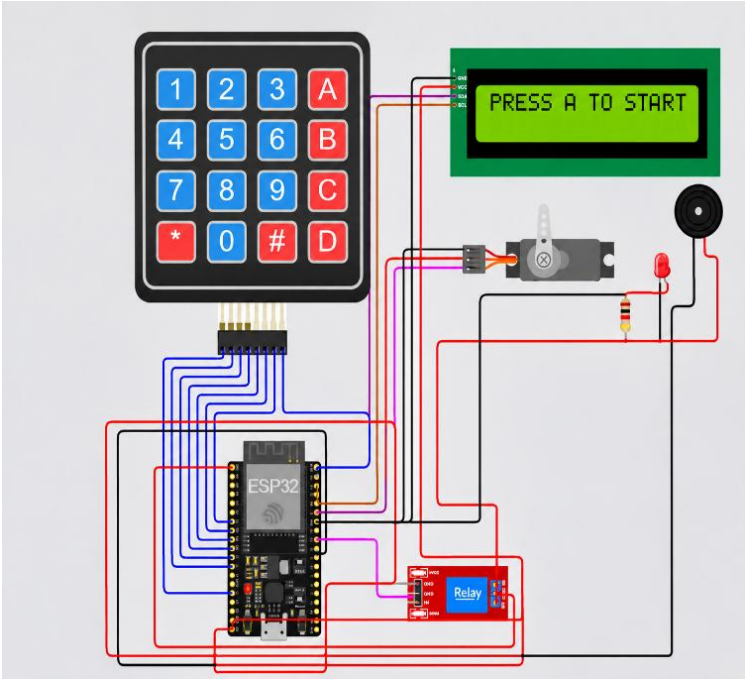


Figure 4.3: Wokwi Simulation Output of the Adaptive Time-Locked Wireless Charging Station

The control circuit and program sequence of the Adaptive Time-Locked Wireless Charging Station were evaluated using the Wokwi online simulator. The simulation was intended to examine the supported control logic and the interaction between the ESP8266 NodeMCU, keypad, LCD, servo motor and relay.

The simulated control sequence includes system initialization, user input, charging-timer operation, relay switching and authentication-based unlocking. When the programmed charging duration expires, the controller switches OFF the relay. The authentication routine permits unlocking when the correct code is entered and maintains the locked condition when an incorrect code is entered.

The Buzzer and SOUND interface must be evaluated according to the simulation capabilities available for these components. If their operation is not represented in the simulation, Buzzer-based phone detection should be verified using the physical hardware.

The simulation results therefore demonstrate only the control functions that were actually implemented and observed in the simulation environment.

COMPONENT	**COST (₹)**
ESP8266	550
16x2 LCD + I2C module	250
4x4 matrix keypad	100
SG90 servo motor	120
5V relay module	80
Wireless charging pad + coil	350
Buzzer, LED, resistors	30
Enclosure / bay door	400
Adapter, wires, PCB	250

Total	**₹ 2130**

Table 4.1: Cost Tabulation of the Adaptive Time-Locked Wireless Charging Station

CHAPTER-5

CONCLUSION AND FUTURE SCOPE

5.1 CONCLUSION

The Adaptive Time-Locked Wireless Charging Station was designed and implemented to combine wireless smartphone charging with physical access protection, user authentication and time-controlled operation. The system uses an ESP8266 NodeMCU as the central controller and integrates a Buzzer with SOUND interface, 4×4 matrix keypad, Mini Servo SG90, relay module, 16×2 LCD, buzzer and wireless charging module.

The Buzzer provides a method of detecting smartphone placement and removal by measuring changes in the Buzzer applied to the charging platform. The keypad allows the user to enter the charging duration and security code. The servo motor controls the compartment's locking mechanism, while the relay controls power supplied to the wireless charging module.

The charging operation is controlled by a programmed countdown timer. When the selected duration expires, the relay switches OFF. The compartment remains locked until the correct authentication code is entered. An incorrect code activates the buzzer and prevents unauthorized unlocking. The LCD provides instructions and status information throughout the operation.

The control logic can be verified through simulation for supported components, while the physical prototype is required to evaluate the actual Buzzer-sound response, wireless charging operation and electromechanical performance. Overall, the proposed system provides a foundation for secure, time-managed smartphone charging in public and semi-public environments.

5.2 SDG MAPPING AND JUSTIFICATION

Goal 9 – Industry, Innovation and Infrastructure: The project supports this goal by integrating the ESP8266 controller, wireless charging module, relay, keypad, display, sensing input and servo-operated lock into one embedded system. The reported prototype tests demonstrated coordinated operation of the main functions, including relay response within approximately 200 ms and servo unlocking within approximately 500 ms after correct authentication.

Goal 11 – Sustainable Cities and Communities: The project is designed for shared public and semi-public settings such as colleges, libraries, airports and railway stations. The test sequence demonstrated that the compartment remains locked during the charging session and does not unlock after an incorrect code; this supports safer and more convenient shared charging facilities, although it does not by itself establish broader city-level sustainability outcomes.

Goal 12 – Responsible Consumption and Production: The user-selected timer and relay-controlled charging supply stop power to the wireless charging module when the chosen charging period expires. The reported tests confirmed relay deactivation at timer expiry. This provides controlled operation and can avoid leaving the charging module powered beyond the selected period; however, the prototype does not directly measure energy savings or reductions in electronic waste.

5.3 FUTURE SCOPE

The system can be improved by introducing IoT-based monitoring using the ESP8266 Wi-Fi capability. A future version could record charging sessions, monitor system status and provide notifications through a connected application.

The authentication mechanism could be enhanced using RFID, biometric authentication or one-time passwords. Multiple charging compartments could also be integrated to support several users.

Further improvements could include more accurate Buzzer-sound calibration, improved power regulation, a dedicated regulated supply for the servo motor and additional wireless charging safety

features. Intelligent charging control and battery-state monitoring could also be investigated in future development.

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